# Study Breaks and Perceived Learning

Exploratory Data Analysis

STA304H5F: Surveys, Sampling and Observational Data
Fall 2026 October 24, 2026

## 1 Study design and data preparation

This report follows the latest group proposal. Its three planned analyses are:

1. **RQ1: Simple linear regression.** Is break count per session associated with agreement that breaks improve efficiency in the next study cycle?
2. **RQ2: Simple linear regression.** Is break duration linearly associated with perceived retention?
3. **RQ3: One-way ANOVA.** Does mean perceived retention differ across break-activity profiles?

The CSV contains 57 records and 13 columns, dated October 5–9, 2026. Consistent with the revised proposal, retention is a next-day self-rating (1–10). Break duration is numeric in minutes; no category midpoints are imputed. Efficiency is agreement with a benefit statement (1–7), rather than measured productivity.

**Sampling.** Recruitment through the course Google Form exchange produced a voluntary-response sample. This report analyzes the screened responses; no random sample of the course was drawn. Self-selection and screening limit generalization to all Fall 2026 STA304H5 students.

**Table 1. Preparation rules and analysis denominators.**

| Issue | Action |
|---|---|
| Unavailable course IDs | Exclude records 8 and 38, whose IDs are missing in the new CSV. Other IDs have not been verified against the course roster. |
| Inconsistent session times | Exclude records 12, 17, 30, 33, 43, 48 and 50: break count times duration meets or exceeds total session duration. |
| Weekly-hour outliers | Set four impossible values (900, 2,800, 600, 600) to missing in this field only. A supplied unresolved range is already missing. |
| Undefined break duration | Set duration to missing for both zero-break respondents, including one reporting 60 minutes despite zero breaks. One additional duration is already missing from an unresolved range; no midpoint is imputed. |
| Analysis sizes | 57 source records; 9 record exclusions; 48 screened records; 46 break takers; 45 complete regression cases; 43 usable weekly-hour values. |

The source CSV is unchanged. Timing flags are conservative analysis exclusions, not proof that the responses are invalid; this screening choice can affect results. Record 25 reports a 780-minute session and 120-minute breaks. It is unusual but internally possible and is retained in the primary fit. Grade values are band proxies, not exact marks. IDs, timestamps and quality notes are administrative fields. Source-record numbers in this report count from 1 after the CSV header. Appendix B visualizes the timing and weekly-hour checks.

## 2 Summary statistics

**Table 2. Numeric summaries of the screened sample.**

| Measure | $n$ | Missing | Mean | Median | SD | IQR | Min | Max |
|---|---|---|---|---|---|---|---|---|
| Study session duration (min) | 48 | 0 | 150.62 | 120.00 | 109.44 | 60.00 | 30.00 | 780.00 |
| Breaks per study session | 48 | 0 | 2.35 | 2.00 | 1.64 | 2.00 | 0.00 | 8.00 |
| Duration of each break (min) | 45 | 3 | 16.89 | 10.00 | 18.81 | 7.50 | 5.00 | 120.00 |
| Next-day retention rating (1-10) | 48 | 0 | 6.15 | 6.00 | 1.40 | 1.00 | 3.00 | 9.00 |
| Weekly study time (hours) | 43 | 5 | 15.81 | 15.00 | 9.61 | 12.00 | 5.00 | 40.00 |
| Prior grade: band proxy (%) | 44 | 4 | 74.77 | 75.00 | 8.49 | 7.50 | 55.00 | 95.00 |
| Break benefit agreement (1-7) | 48 | 0 | 5.73 | 6.00 | 1.16 | 1.00 | 2.00 | 7.00 |

Generated with `datawizard::describe_distribution()` from the easystats ecosystem. SD is the sample SD; IQR uses datawizard's R type-6 quantiles. The three unavailable break durations comprise two structurally undefined no-break cases and one unresolved range. Grade proxies and ordinal ratings are summarized numerically for completeness, without assuming exact interval measurement. Administrative fields are not substantive variables.

Median session length is 120 minutes, with two breaks per session. Break duration has median 10 minutes, mean 16.89 and SD 18.81, indicating a long upper tail. Retention averages 6.15 (SD 1.40). Valid weekly study time averages 15.81 hours ($n = 43$). Similar means and medians alone do not establish normality.

**Table 3a. Prior course grade band.**

| Category | $n$ | % of 48 |
|---|---|---|
| 50-59% | 1 | 2.1 |
| 60-69% | 10 | 20.8 |
| 70-79% | 25 | 52.1 |
| 80-89% | 5 | 10.4 |
| 90-100% | 3 | 6.2 |
| Missing / unavailable | 4 | 8.3 |

**Table 3b. Break benefit agreement.**

| Category | $n$ | % of 48 |
|---|---|---|
| Strongly disagree | 0 | 0.0 |
| Disagree | 2 | 4.2 |
| Slightly disagree | 0 | 0.0 |
| Neutral | 4 | 8.3 |
| Slightly agree | 8 | 16.7 |
| Agree | 23 | 47.9 |
| Strongly agree | 11 | 22.9 |
| Missing / unavailable | 0 | 0.0 |

The 70–79% grade band was most common (52.1%). Forty-two respondents (87.5%) at least slightly agree that breaks help efficiency. Four prior grades are unavailable; the recorded data do not establish why. Percentages use all 48 screened respondents, including missing cases; totals may differ from 100% due to rounding.

**Table 3c. Recorded year/attention-check item responses.**

| Recorded code | $n$ | % of 48 |
|---|---|---|
| 1 | 5 | 10.4 |
| 2 | 0 | 0.0 |
| 3 | 29 | 60.4 |
| 4 | 14 | 29.2 |
| 5 | 0 | 0.0 |
| Missing | 0 | 0.0 |

Counts describe recorded responses, not verified years of study. Interpretation as an attention check requires confirmation of the administered prompt.

## 3 Response distributions and visual exploration

**Table 4a. Reported break activities.**

| Activity | $n$ | % of 48 |
|---|---|---|
| Eating | 17 | 35.4 |
| Exercise | 9 | 18.8 |
| Gaming | 17 | 35.4 |
| Napping | 11 | 22.9 |
| none | 1 | 2.1 |
| Instagram write-in | 1 | 2.1 |
| Scrolling | 1 | 2.1 |
| Scrolling Phone | 1 | 2.1 |
| Social | 37 | 77.1 |
| Space out | 1 | 2.1 |
| Youtube | 1 | 2.1 |

**Table 4b. Reported study locations.**

| Location | $n$ | % of 48 |
|---|---|---|
| Cafe | 4 | 8.3 |
| Home | 41 | 85.4 |
| Lecture hall | 6 | 12.5 |
| Library | 35 | 72.9 |
| Other campus building | 2 | 4.2 |
| Outdoors | 1 | 2.1 |
| Public transit | 3 | 6.2 |
| Missing / unusable | 1 | 2.1 |

Multiple selections are counted once per respondent per label; percentages can exceed 100%. All selections are retained, including those exceeding the requested two. "Social" remains the source label without an assumed activity type. The Instagram write-in is shortened for display. Location splitting preserves commas inside parentheses, and campus-building aliases are combined. Activities have no missing responses; one location is unusable.

"Social" is selected by 77.1% of respondents; home (85.4%) and library (72.9%) are the most frequently reported locations. These categories overlap.

**Table 4c. Distribution of perceived next-day retention.**

| Rating | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Count | 0 | 0 | 3 | 4 | 4 | 17 | 13 | 6 | 1 | 0 |
| % of 48 | 0 | 0 | 6.2 | 8.3 | 8.3 | 35.4 | 27.1 | 12.5 | 2.1 | 0 |

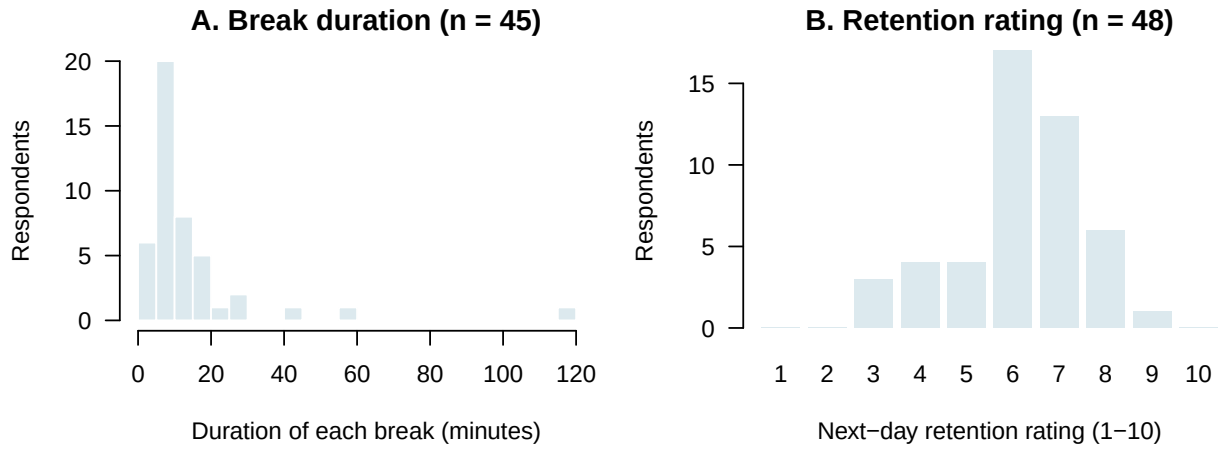



**Figure 1. Distributions of the regression variables.** A: observed break duration for the 45 model cases; the isolated 120-minute value creates a long right tail. B: retention for all 48 screened respondents, with 62.5% rating 6 or 7. Retention is bounded and discrete; residual diagnostics, rather than normality of the predictor, determine the suitability of normal-error inference.

# 4 Planned tests and RQ1: Simple linear regression

The three proposal analyses use $\alpha = 0.05$. Raw and Holm-adjusted $p$-values are reported for this family of three tests. Assumption limitations remain relevant.

**Table 5. Tests specified in the revised proposal.**

| Question | Test | $n$ | Statistic | Raw $p$ | Holm $p$ |
|---|---|---|---|---|---|
| RQ1 | Simple linear regression | 48 | 1.933 | .0594 | .1783 |
| RQ2 | Simple linear regression | 45 | −0.070 | .9445 | .9445 |
| RQ3 | One-way ANOVA | 46 | .977 | .4128 | .8257 |

Statistics are slope $t$ with 46 and 43 df, and ANOVA $F$ with (3,42) df.

**Model and result.** Regress the seven-point benefit-agreement score on the numerical break count per session; test $H_0 : \beta_1 = 0$ against $H_A : \beta_1 \neq 0$. The estimated slope is 0.1938 points per additional break (95% CI: −0.0080, 0.3956), with $R^2 = 0.0751$. The positive fitted association is not significant at 0.05, before or after adjustment. It does not measure an actual gain in efficiency.

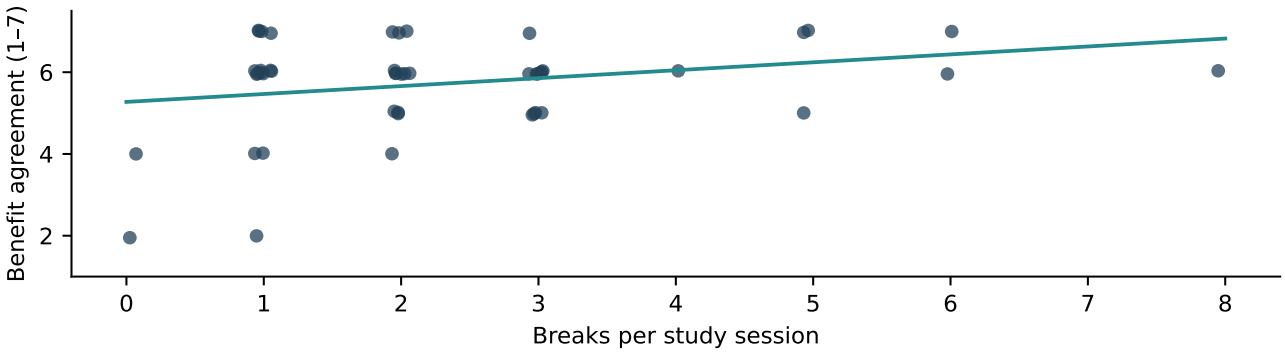



**Figure 1a.** Break count and benefit agreement. Jitter separates overlapping points; the fitted line uses unjittered observations.

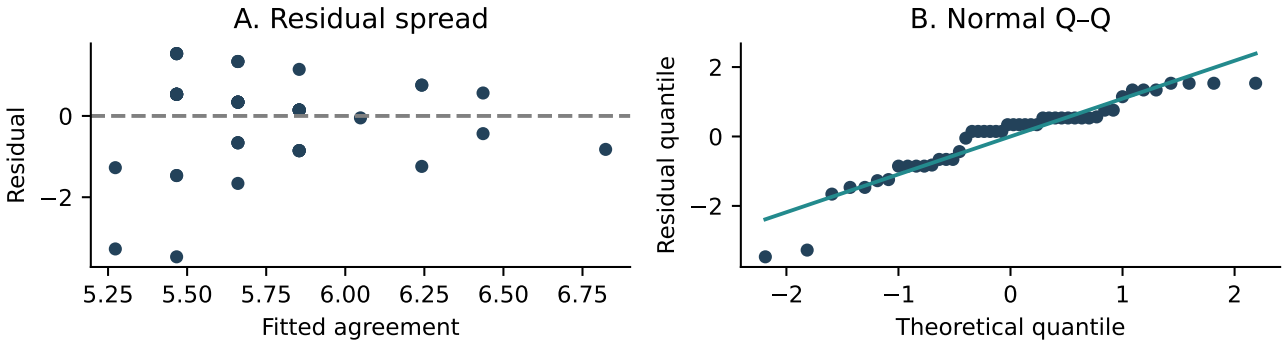



**Figure 1b.** RQ1 residual spread and normal Q–Q plot. Uneven spread and departures from the reference line limit normal, constant-variance inference.

**Assumptions.** Shapiro–Wilk $p = 0.00085$ and the variance score test $p = 0.00484$ indicate departures from normality and constant variance. The bounded, ordinal response also limits the linear approximation. Independence cannot be established from these plots. Interpret the classical interval and $p$-value cautiously.

## 5 RQ2: Simple linear regression

Let $X_i$ be break duration in minutes and $Y_i$ the next-day retention rating. For the 45 complete break takers, fit

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \qquad H_0 : \beta_1 = 0 \quad \text{versus} \quad H_A : \beta_1 \neq 0.$$

The primary test is two-sided at $\alpha = 0.05$. It is one of the three exploratory tests planned in the proposal. The model treats the rating scale as approximately numeric.

**Table 7. Ordinary least-squares coefficients and conventional 95% CIs.**

| Term | Estimate | SE | $t$ | $p$ | CI lower | CI upper |
|---|---|---|---|---|---|---|
| Intercept | 6.21302 | 0.27670 | 22.45421 | $< 0.00001$ | 5.65501 | 6.77103 |
| Break duration (min) | -0.00077 | 0.01101 | -0.07000 | 0.94452 | -0.02298 | 0.02144 |

$n = 45$; residual df $= 43$; $R^2 = 0.000114$; adjusted $R^2 = -0.02314$; residual standard error $= 1.374$ rating points. Overall $F(1, 43) = 0.0049$, with the same $p$-value as the slope test. Conventional inference is approximate given the diagnostics in Section 7.

The fitted equation is

$$\widehat{\text{retention}} = 6.21302 + (-0.000771) \times \text{break minutes}.$$

The slope is -0.000771 rating points per minute (95% CI $[-0.02298, 0.02144]$), $t(43) = -0.070$, $p = 0.9445$. A 10-minute longer break corresponds to an estimated -0.0077 point difference (95% CI $[-0.2298, 0.2144]$). The fitted direction is slightly negative, but the slope is not statistically significant and the explained variation is less than 0.02%. This does not establish that the association is zero.

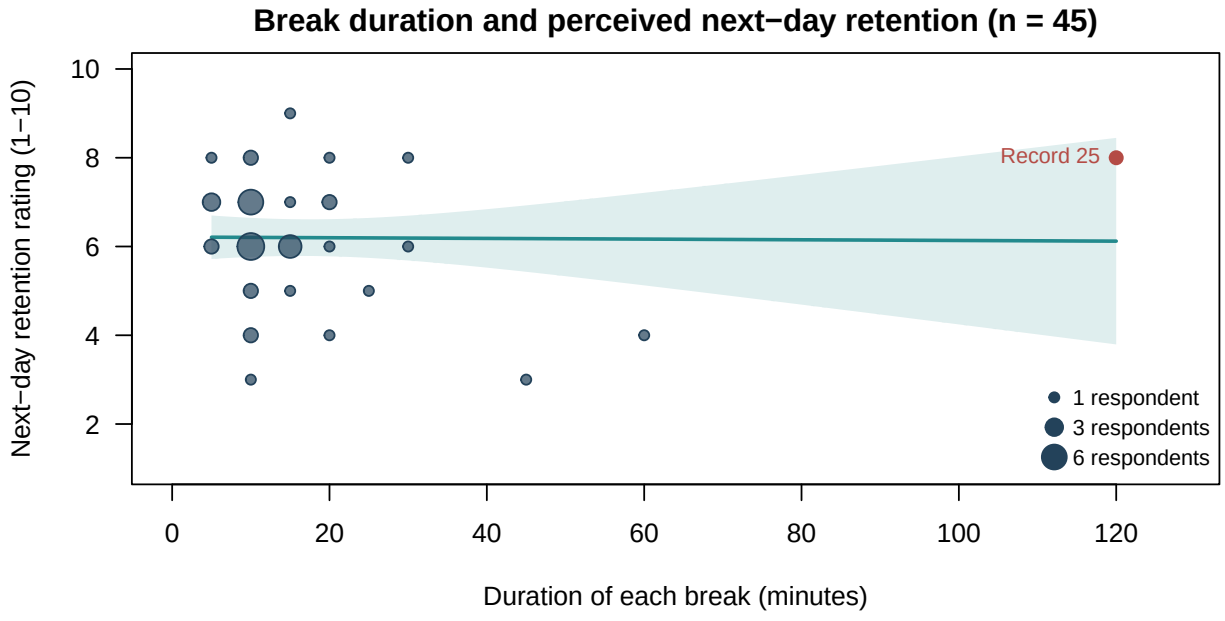



**Figure 2. Retention versus duration and fitted regression.** Point area represents the number of identical responses. The teal line is the fitted conditional mean; shading is its conventional pointwise 95% confidence band, not an interval for individual responses. The sparse upper-duration range and record 25 make the near-flat line sensitive to influence.

The intercept predicts a rating at zero minutes, outside the observed 5–120 minute range. It is not a comparison with students taking no breaks, who are excluded from this model because their break duration is undefined.

# 6 RQ3: One-way ANOVA of activity profiles

The revised proposal compares four mutually exclusive exercise/gaming profiles: exercise without gaming, gaming without exercise, neither, and both. This definition handles the multiple-selection activity question without duplicating respondents across ANOVA groups. Other activities may co-occur in every group, so the comparison concerns these profiles rather than every individual activity type.

All 46 screened break takers enter this comparison, including one with missing break duration. Each contributes once. "Only" below refers to exercise versus gaming, not to the absence of all other activities. The ambiguous label "Social" is not recoded or used to assign these profiles.

**Table 8. Perceived next-day retention by exclusive activity profile.**

| Profile | $n$ | Mean | SD | Median |
|---|---|---|---|---|
| Exercise only | 5 | 6.20 | 0.45 | 6.00 |
| Gaming only | 12 | 5.75 | 1.54 | 6.00 |
| Neither | 25 | 6.32 | 1.35 | 7.00 |
| Both | 4 | 7.00 | 1.41 | 7.50 |

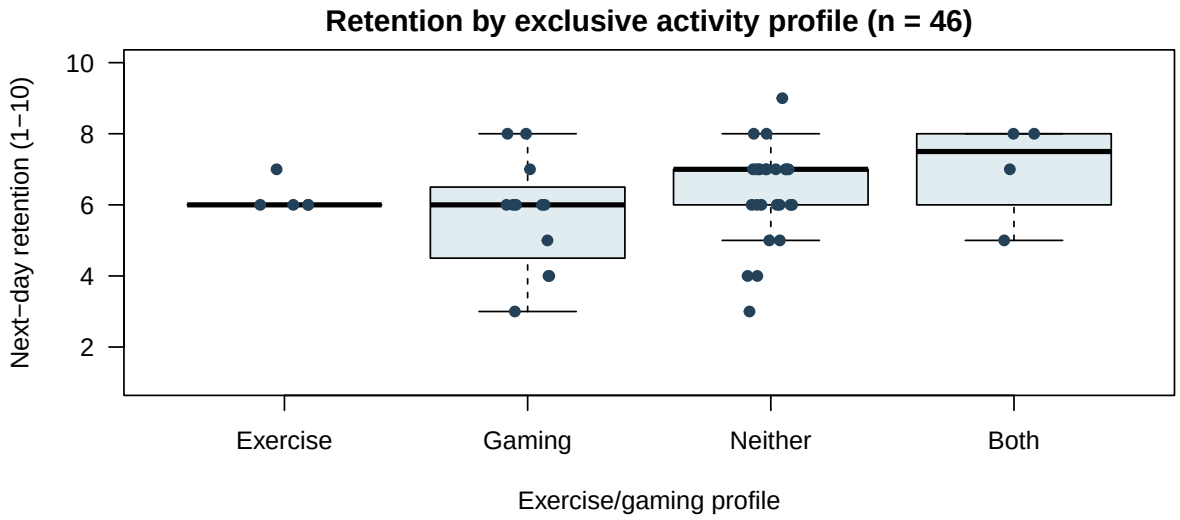



**Figure 3. Retention across exclusive activity profiles.** Boxes show medians and quartiles; points show individual observations with horizontal jitter only. Exercise and gaming labels indicate the corresponding exclusive profiles. The "both" group has the highest mean but only four respondents, and the distributions overlap considerably.

Test $H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$ against at least one unequal mean. The planned one-way ANOVA gives $F(3, 42) = 0.977$, $p = 0.4128$, with $\eta^2 = 0.065$. It does not reject equality at 0.05. The higher observed mean in the "both" profile is not a statistically established difference. No post-hoc comparisons or additional rank-based or unequal-variance tests are performed. This comparison does not isolate the effect of a single activity.

# 7 Model assumption and influence checks

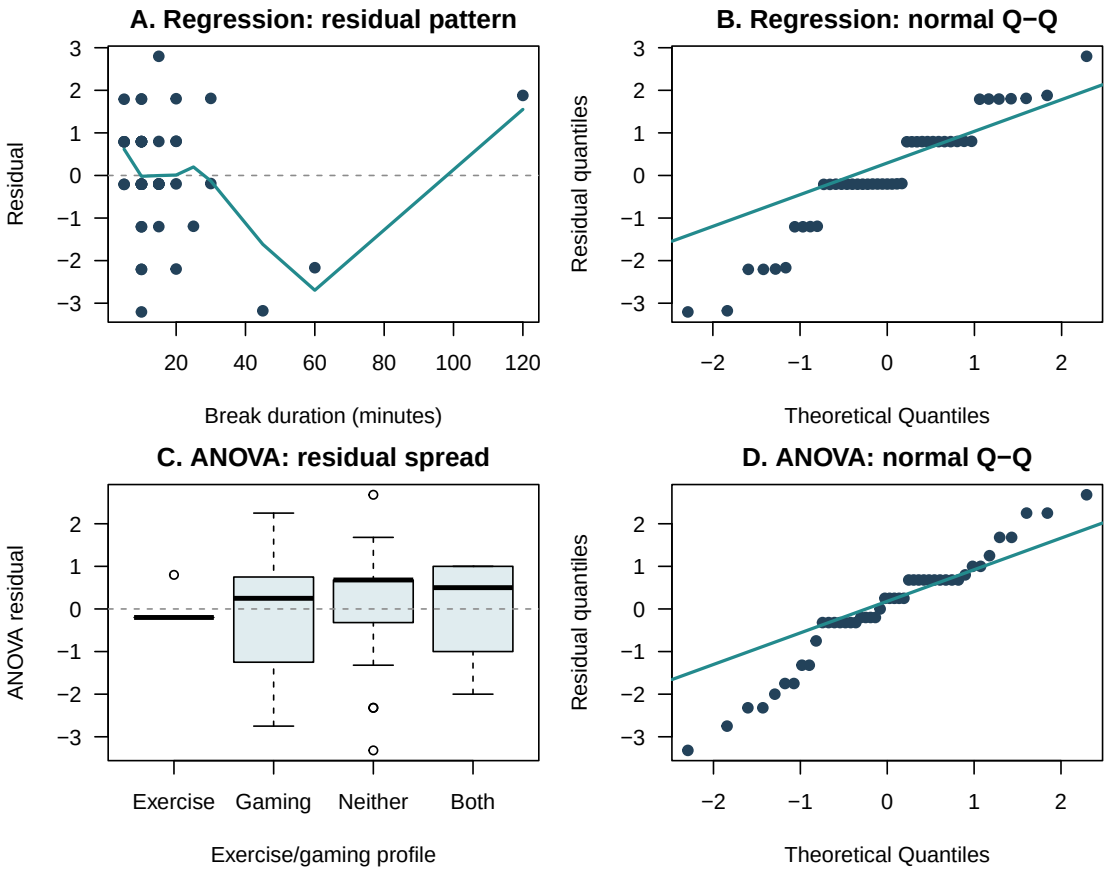



**Figure 4. Diagnostics for the planned regression and ANOVA.** A: regression residual pattern and spread against duration. B: its normal Q–Q plot. C: ANOVA residual spread across profiles. D: ANOVA normal Q–Q plot. Tied ratings and sparse long breaks limit the plots' resolution.

**Table 9. Diagnostic tests supplementing the plots.**

| Model | Check | $p$ |
|---|---|---|
| Regression | Shapiro-Wilk residual normality | 0.00975 |
| Regression | Variance score test | 0.04998 |
| ANOVA | Shapiro-Wilk residual normality | 0.06311 |
| ANOVA | Median-centered Levene test | 0.43691 |

**Regression.** The residual trend is not uniformly flat, Q–Q tails depart from normality, and Shapiro–Wilk rejects ($p = 0.00975$). The variance score test is just below 0.05 ($p = 0.04998$), giving borderline evidence against equal variance. Conventional regression inference is consequently approximate. The retained 120-minute response (record 25) has leverage 0.705 and Cook's distance 7.60, indicating strong influence. It is not deleted to change significance, and no alternative regression or deletion test is added.

**ANOVA.** Neither normality nor median-centered Levene's test rejects at 0.05. This does not prove the assumptions, especially with groups of four and five. Independence is assumed from individual survey responses and exclusive profile membership; neither diagnostic plots nor ID screening can verify it. The course permits reporting violated assumptions while proceeding with the planned analysis; the results above are interpreted with those limitations.

# 8 Interpretation, alignment and limitations

**RQ1.** The fitted association is positive (0.1938 agreement points per additional break), but not significant (raw $p = 0.0594$; Holm $p = 0.1783$). Residual normality and variance departures further limit the classical inference.

**RQ2.** The fitted duration slope is near zero and not significant ($p = 0.9445$). Residual departures and one highly influential response limit inference. The result does not prove that duration has no relationship with retention and does not identify an optimal break length.

**RQ3.** The planned ANOVA does not find a significant mean difference across the four exclusive exercise/gaming profiles ($p = 0.4128$). The largest observed group mean is descriptive and is based on only four responses.

**Alignment with the revised proposal.** RQ1 and RQ2 use simple linear regression; RQ3 uses one-way ANOVA of exclusive exercise/gaming profiles. Raw and Holm-adjusted $p$-values are reported. All three adjusted results exceed 0.05. Diagnostic checks support the EDA. The proposal's sensitivity analyses remain follow-up work, not completed results in this report.

**Limitations.**

1. **Sampling and eligibility.** Participation was voluntary. Self-selection and screening may introduce bias, and remaining course IDs are not roster-verified. Results do not represent a validated random sample of the full course.
2. **Preparation.** The missing-ID and timing exclusions, structural missing durations for zero-break cases, and variable-level weekly-hour exclusions are transparent preparation choices. Flags alone do not prove invalidity. Unresolved ranges remain missing. These choices can affect the analysis sample.
3. **Measurement.** Retention is a bounded ordinal self-rating, not an objective test score; agreement is an attitude. Grades are band proxies. Treating the ratings numerically is an approximation.
4. **Year/attention-check item.** Five of 48 respondents selected code 1. If the administered prompt instructed respondents to select "First Year, the other 43 responses did not follow that instruction. This interpretation requires verification against the actual form. These responses are not used as verified years of study or as an additional exclusion rule.
5. **Confounding and precision.** Session length, study load and personal differences are not adjusted for. Small activity groups, co-occurring choices and influential long breaks limit inference. None of the analyses supports a causal learning-improvement claim.

**Reproducibility.** Appendix A provides the corrected R script, using `datawizard` for summaries. The supplied `report.tex` recompiles this PDF from preserved pages and editable corrections. Corrections were independently checked in Python; the revised R script has not been executed in this environment.

**References.** Revised group proposal (`proposal_reference.txt`); STA304 Exploratory Data Analysis Instructions (2026). Patil et al. (2022), *datawizard: An R Package for Easy Data Preparation and Statistical Transformations*, JOSS 7(78), 4684, doi:10.21105/joss.04684. The R script exports package citation and session information when run.

# A Complete reproducible R code

Run `Rscript analysis.R` from this folder after installing `datawizard`, `bayestestR`, `car` and `jsonlite`. The script reads `survey_source.csv` without modifying it. It writes tables, plots and session information. The revised script requires execution in R before submission as fully run analysis.

```
# STA304 EDA. Run Rscript analysis.R from the submission folder.
# Input: unchanged copy of the new 57-record CSV and its codebook.
# This script computes statistics, preparation and graphics in R.
options(stringsAsFactors = FALSE, width = 78)
needed <- c("datawizard", "bayestestR", "car", "jsonlite")
if (!all(vapply(needed, requireNamespace, logical(1), quietly = TRUE))) {
  stop("Install the required packages: ", paste(needed, collapse = ", "))
}
dir.create("tables", showWarnings = FALSE)
dir.create("figures", showWarnings = FALSE)
raw <- read.csv("survey_source.csv", na.strings = "", check.names = FALSE,
  fileEncoding = "UTF-8-BOM", colClasses = c(studentID = "character"))
raw$record <- seq_len(nrow(raw))
raw$bad_id <- is.na(raw$studentID) | !nzchar(trimws(raw$studentID))
total <- with(raw, num_breaks * break_duration_min)
raw$bad_time <- !is.na(total) & total >= raw$study_session_min
raw$bad_week <- !is.na(raw$weekly_study_hours) &
  (raw$weekly_study_hours > 168 | raw$weekly_study_hours < 0)
raw$excluded <- raw$bad_id | raw$bad_time
d <- raw[!raw$excluded, ]
d$weekly_study_hours[d$bad_week] <- NA_real_
# No-break respondents have no defined duration of an actual break.
# Do not infer a duration of zero from the contradictory positive answer.
d$break_duration_min[d$num_breaks == 0] <- NA_real_
split_multi <- function(x) {
  lapply(x, function(z) if (is.na(z)) character() else
    strsplit(z, ",\\s*(?![^()]*\\))", perl = TRUE)[[1]])
}
activities <- split_multi(d$break_activities)
has <- function(label) vapply(activities, function(z) label %in% z, logical(1))
labels <- c(study_session_min = "Study session duration (min)",
  num_breaks = "Breaks per study session",
  break_duration_min = "Duration of each break (min)",
  memory_retention = "Next-day retention rating (1-10)",
  weekly_study_hours = "Weekly study time (hours)",
  prior_grade_pct = "Prior grade: band proxy (%)",
  break_helps_efficiency = "Break benefit agreement (1-7)")
ds <- as.data.frame(datawizard::describe_distribution(d[names(labels)],
  centrality = c("mean", "median")))
write.csv(ds, "tables/easystats_summary.csv", row.names = FALSE)
num <- data.frame(Variable = unname(labels[ds$Variable]), n = ds$n,
  Missing = ds$n_Missing, Mean = ds$Mean, Median = ds$Median,
  SD = ds$SD, IQR = ds$IQR,
  Min = vapply(d[ds$Variable], min, numeric(1), na.rm = TRUE),
  Max = vapply(d[ds$Variable], max, numeric(1), na.rm = TRUE))
freq <- function(x, levels, labels = levels) {
  z <- table(factor(x, levels = levels), useNA = "always")
  data.frame(Category = c(labels, "Missing / unavailable"),
    n = as.integer(z), Percent = 100 * as.integer(z) / length(x))
}
```

```
year <- freq(d$year_of_study, 1:5, paste("Recorded code", 1:5))
grade <- freq(d$prior_grade_pct, c(55, 65, 75, 85, 95),
  c("50-59%", "60-69%", "70-79%", "80-89%", "90-100%"))
agree <- freq(d$break_helps_efficiency, 1:7,
  c("Strongly disagree", "Disagree", "Slightly disagree", "Neutral",
    "Slightly agree", "Agree", "Strongly agree"))
memory <- freq(d$memory_retention, 1:10, as.character(1:10))
al <- sort(unique(unlist(activities)))
avalues <- c(Intercept = sprintf("%.5f", coef(ols)[1]),
  Slope = sprintf("%.6f", coef(ols)[2]),
  SlopeP = sprintf("%.4f", coeff$p[2]),
  SlopeT = sprintf("%.3f", coeff$t[2]),
  SlopeLo = sprintf("%.5f", ci[2, 1]),
  SlopeHi = sprintf("%.5f", ci[2, 2]),
  RSquared = sprintf("%.6f", sm$r.squared),
  AdjRSquared = sprintf("%.5f", sm$adj.r.squared),
  ResidualSE = sprintf("%.3f", sm$sigma),
  FStat = sprintf("%.4f", unname(sm$fstatistic[1])),
  TenEffect = sprintf("%.4f", 10 * coef(ols)[2]),
  TenLo = sprintf("%.4f", 10 * ci[2, 1]),
  TenHi = sprintf("%.4f", 10 * ci[2, 2]),
  ShapiroP = sprintf("%.5f", sh_ols$p.value),
  NCVP = sprintf("%.5f", ncv$p),
  MaxCook = sprintf("%.2f", max(cooks)),
  MaxHat = sprintf("%.3f", max(hatvalues(ols))),
  RQOneSlope = sprintf("%.5f", coef(rq1)[2]),
  RQOneP = sprintf("%.4f", rq1_sm$coefficients[2, 4]),
  RQOneLo = sprintf("%.5f", rq1_ci[2, 1]),
  RQOneHi = sprintf("%.5f", rq1_ci[2, 2]),
  AnovaF = sprintf("%.3f", avtab[1, "F value"]),
  AnovaP = sprintf("%.4f", avtab[1, "Pr(>F)"]),
  EtaSquared = sprintf("%.3f", avtab[1, "Sum Sq"] / sum(avtab[, "Sum Sq"])))
names(values) <- sub("\\..*$", "", names(values))
writeLines(paste0("\\newcommand{\\", names(values), "}{", values, "}"),
  "tables/report_values.tex")
navy <- "#23425A"; teal <- "#238A8D"; red <- "#B44B45"
pdf("figures/rq1_results.pdf", width = 7.2, height = 2.5)
par(mar = c(4, 4, 1, 1), cex = .85, las = 1)
set.seed(304)
plot(jitter(d$num_breaks, amount = .07),
  jitter(d$break_helps_efficiency, amount = .05), pch = 19, col = navy,
  xlab = "Breaks per study session", ylab = "Benefit agreement (1-7)")
abline(rq1, col = teal, lwd = 2)
dev.off()
pdf("figures/rq1_diagnostics.pdf", width = 7.2, height = 2.5)
par(mfrow = c(1, 2), mar = c(4, 4, 2, 1), cex = .8, las = 1)
plot(fitted(rq1), residuals(rq1), pch = 19, col = navy,
  xlab = "Fitted agreement", ylab = "Residual", main = "RQ1: residual spread")
abline(h = 0, lty = 2, col = "grey55")
lines(lowess(fitted(rq1), residuals(rq1)), col = teal, lwd = 2)
qqnorm(residuals(rq1), pch = 19, col = navy, main = "RQ1: normal Q-Q")
qqline(residuals(rq1), col = teal, lwd = 2)
dev.off()
pdf("figures/descriptive_distributions.pdf", width = 7.2, height = 2.65)
par(mfrow = c(1, 2), mar = c(4.1, 3.9, 2.2, .7), cex = .82, las = 1)
hist(bt$break_duration_min, breaks = seq(0, 125, 5), col = "#DCE9EE",
```

```
  border = "white", main = "A. Break duration (n = 45)",
  xlab = "Duration of each break (minutes)", ylab = "Respondents")
barplot(as.integer(table(factor(d$memory_retention, levels = 1:10))),
  names.arg = 1:10, col = "#DCE9EE", border = NA,
  main = "B. Retention rating (n = 48)",
  xlab = "Next-day retention rating (1-10)", ylab = "Respondents")
dev.off()
pdf("figures/research_results.pdf", width = 7.2, height = 3.65)
par(mar = c(4.2, 4.2, 2.1, .8), cex = .92, las = 1)
grid <- data.frame(break_duration_min = seq(min(bt$break_duration_min),
  max(bt$break_duration_min), length.out = 250))
prediction <- predict(ols, grid, interval = "confidence")
plot(NA, xlim = c(0, 128), ylim = c(1, 10),
  xlab = "Duration of each break (minutes)",
  ylab = "Next-day retention rating (1-10)",
  main = "Break duration and perceived next-day retention (n = 45)")
polygon(c(grid$break_duration_min, rev(grid$break_duration_min)),
  c(prediction[, "lwr"], rev(prediction[, "upr"])),
  col = adjustcolor(teal, alpha.f = .15), border = NA)
lines(grid$break_duration_min, prediction[, "fit"], col = teal, lwd = 2)
points_n <- aggregate(rep(1, nrow(bt)),
  list(Minutes = bt$break_duration_min, Rating = bt$memory_retention), sum)
points(points_n$Minutes, points_n$Rating, pch = 21,
  bg = adjustcolor(navy, alpha.f = .7), col = navy,
  cex = .85 * sqrt(points_n$x))
im <- which.max(cooks)
points(bt$break_duration_min[im], bt$memory_retention[im],
  pch = 21, col = red, bg = red, cex = 1.1)
text(bt$break_duration_min[im], bt$memory_retention[im],
  labels = "Record 25", pos = 2, col = red, cex = .83)
legend("bottomright", c("1 respondent", "3 respondents", "6 respondents"),
  pch = 21, pt.bg = navy, col = navy, pt.cex = .85 * sqrt(c(1, 3, 6)),
  bty = "n", cex = .78, y.intersp = 1.2)
dev.off()
short <- c("Exercise", "Gaming", "Neither", "Both")
pdf("figures/activity_comparison.pdf", width = 7.2, height = 3.2)
par(mar = c(4.2, 4, 2, .8), cex = .9, las = 1)
boxplot(memory_retention ~ activity_profile, data = ad, names = short,
  col = "#E0ECEF", xlab = "Exercise/gaming profile",
  ylab = "Next-day retention (1-10)", ylim = c(1, 10),
  main = "Retention by exclusive activity profile (n = 46)", outline = FALSE)
set.seed(304)
stripchart(memory_retention ~ activity_profile, data = ad,
  method = "jitter", vertical = TRUE, add = TRUE, pch = 19, col = navy)
dev.off()
pdf("figures/model_diagnostics.pdf", width = 7.2, height = 5.6)
par(mfrow = c(2, 2), mar = c(4, 4, 2.1, 1), cex = .85, las = 1)
plot(bt$break_duration_min, residuals(ols), pch = 19, col = navy,
  xlab = "Break duration (minutes)", ylab = "Residual",
  main = "A. Regression: residual pattern")
abline(h = 0, col = "grey55", lty = 2)
lines(lowess(bt$break_duration_min, residuals(ols)), col = teal, lwd = 2)
qqnorm(residuals(ols), pch = 19, col = navy,
  main = "B. Regression: normal Q-Q", ylab = "Residual quantiles")
qqline(residuals(ols), col = teal, lwd = 2)
boxplot(residuals(av) ~ ad$activity_profile, names = short, col = "#E0ECEF",
```

```
  xlab = "Exercise/gaming profile", ylab = "ANOVA residual",
  main = "C. ANOVA: residual spread")
abline(h = 0, col = "grey55", lty = 2)
qqnorm(residuals(av), pch = 19, col = navy,
  main = "D. ANOVA: normal Q-Q", ylab = "Residual quantiles")
qqline(residuals(av), col = teal, lwd = 2)
dev.off()
pdf("figures/data_quality.pdf", width = 7.2, height = 3.35)
par(mfrow = c(1, 2), mar = c(4, 4, 2.2, 1), cex = .84, las = 1)
ok <- !is.na(total) & total > 0
plot(raw$study_session_min[ok], total[ok], log = "xy", pch = 19,
  col = ifelse(raw$bad_time[ok], red, navy),
  xlab = "Session minutes (log scale)",
  ylab = "Total break minutes (log scale)", main = "A. Session-time consistency")
abline(0, 1, col = "grey55", lty = 2)
extreme <- which(raw$bad_time & total > 1000)
text(raw$study_session_min[extreme], total[extreme],
  labels = paste("Record", extreme), pos = 2, cex = .8)
wk <- !is.na(raw$weekly_study_hours) & raw$weekly_study_hours > 0
plot(raw$record[wk], raw$weekly_study_hours[wk], log = "y", pch = 19,
  col = ifelse(raw$bad_week[wk], red, navy), xlab = "Source record number",
  ylab = "Weekly study time (hours; log scale)", main = "B. Weekly-time outliers")
abline(h = 168, lty = 2, col = red)
text(18, 168, "168 hours/week", pos = 3, cex = .8, col = red)
dev.off()
result <- list(method_reference = "proposal_reference.txt",
  raw_n = nrow(raw), screened_n = nrow(d),
  break_takers_n = nrow(breakers), regression_n = nrow(bt), anova_n = nrow(ad),
  bad_id_records = raw$record[raw$bad_id],
  timing_records = raw$record[raw$bad_time],
  weekly_outlier_records = raw$record[raw$bad_week],
  numeric_summary = num, coefficients = coeff, model_fit = fit,
  primary_tests = tests, activity_groups = groups, assumptions = checks,
  rq1 = list(coefficients = rq1_coeff, r_squared = rq1_sm$r.squared,
    residual_df = df.residual(rq1), shapiro_p = rq1_shapiro$p.value,
    variance_score_p = rq1_ncv$p),
  anova = list(df1 = avtab[1, "Df"], df2 = avtab[2, "Df"],
    eta_squared = avtab[1, "Sum Sq"] / sum(avtab[, "Sum Sq"])),
  influence = list(max_cook_record = bt$record[im],
    max_cook = max(cooks), max_leverage = max(hatvalues(ols))))
jsonlite::write_json(result, "results.json", auto_unbox = TRUE,
  pretty = TRUE, digits = 15)
capture.output(sessionInfo(), file = "session_info.txt")
capture.output(citation("datawizard"), file = "tables/datawizard_citation.txt")
```

# B Source-data consistency and outlier plots

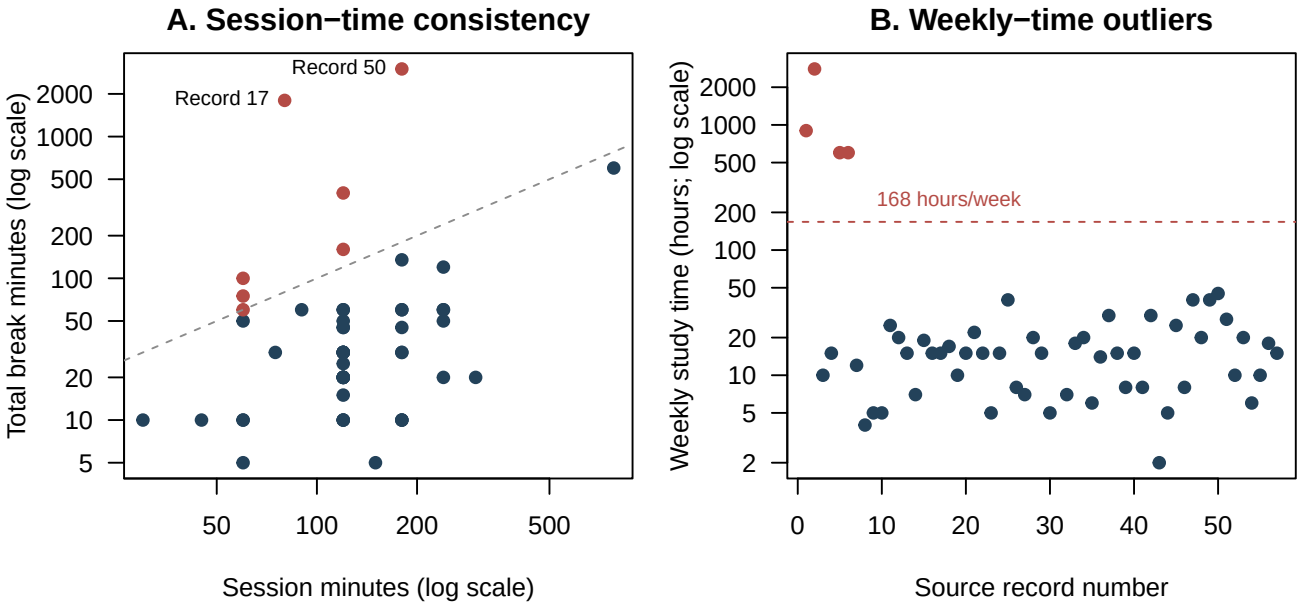



**Figure B1. Source-data consistency checks.** A: the 54 positive, observed total break times against session duration on log scales. The seven red points meet or exceed the equality line and are excluded by the timing screen. Two no-break zero totals and one missing duration are not shown on the log plot. Only the two largest inconsistent totals are labelled. B: the 56 observed weekly-hour entries, including four above the 168-hour limit. Those four fields are set to missing, without assuming a unit conversion. Source-record numbers start from 1 after the CSV header.